







Adaptive Aquaculture Water Quality Monitoring via Zebrafish Using Progressive Hybrid FQL-DQN with N3IWF on TinyML Embedded Platform

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“MCU-Based Wi-Fi Repeater with External Antenna Integration for Optimized IoT Connectivity”

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OUTLINE



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LITERATURE REVIEW

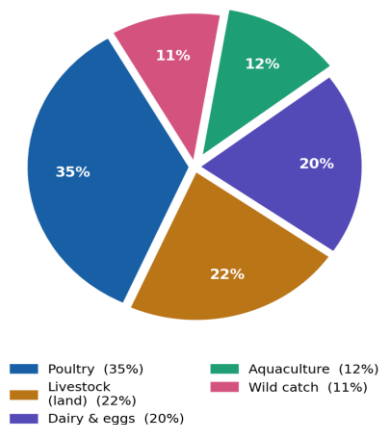
RESEARCH GAP

METHODOLOGY

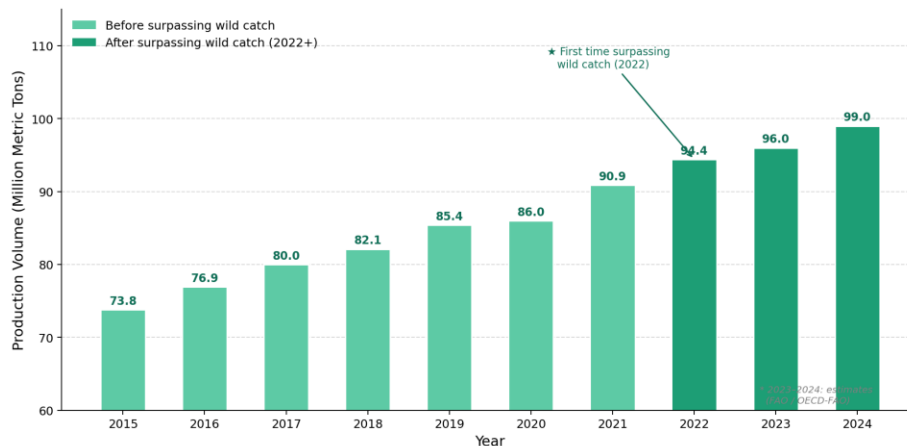


Background

Animal Protein Consumption by Source (%)
Data: 2022



Global Aquaculture Production Trend
2015-2024 (Million Metric Tons)



Aquaculture is a major source of global animal protein, facing growing demand



Taiwan 2023: production value **NT\$40.2** billion → **41%** of total national fisheries output

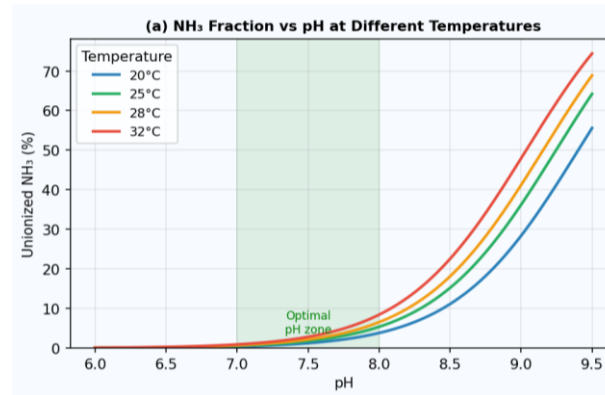
- Global aquaculture loses >US\$8 billion due to water quality-related fish-killing events, with Asia bearing the highest economic impact (Mehdizadeh Allaf et al., 2024)
- NH_3 toxicity causes sudden mass mortality in aquaculture — driven silently by pH and temperature shifts, often undetected until catastrophic fish loss occurs (Kubitza, 2017)



Background

NH₃ Toxicity in Aquaculture & Zebrafish as Biological Monitoring Model

pH and temperature are the most monitored parameters in IoT aquaculture studies (98% and 93%)



Zebrafish (Danio rerio)



Why Zebrafish?

Standard biological model in aquatic toxicology (OECD). High NH₃ sensitivity makes zebrafish ideal as monitoring proxy.

Optimal Parameters

pH: 6.5–8.5

Temp: 25–28°C

NH₃: <0.02 mg/L

When pH 8.5 + T 30°C, NH₃ becomes 5x more toxic than safe limit yet fish appear normal until sudden mass death occurs

NH₃ is fully derivable from pH + T via Henderson-Hasselbalch
— no ammonia sensor needed

NH₃ risk derived from pH + T via Henderson-Hasselbalch | Zebrafish (*Danio rerio*)
— OECD sentinel species | pH 6.5–8.5, T 25–28°C



Background



9 Aquaculture Farm Sites - Taixi Township, Yunlin, Taiwan

Distributed Farm Sites

9 fishponds spread across a large area - each requiring independent real-time water quality monitoring

Local Edge Network

N3IWF connects all monitoring nodes within one local Wi-Fi network - no internet required, low latency guaranteed

Scalable Monitoring

Each Pico WH node monitors pH and temperature independently - all data aggregated at N3IWF for unified dashboard view

N3IWF Local Edge Services enables all farm sites to be monitored through one local network - secure, low-latency, and internet-independent



Background

Existing Methods

Commercial IoT Monitoring

Multi-sensor (pH, DO, ORP, EC, Temp), comprehensive sensing but no adaptive control loop

Rule-Based Control

Safe but static, cannot adapt to dynamic water quality conditions

Fuzzy Logic

Fixed membership functions, cannot learn or adapt from field data over time

ML / DL Systems

Accurate but cloud-dependent, impractical for remote farm sites

Deep RL Systems

Adaptive but server-grade hardware required, not embeddable

VS

This Research

Progressive Hybrid Control

Rule-Based → FQL → DQN: safe, adaptive, and data-efficient

NH₃-Aware Reward Function

Biology-based reward derived directly from pH and temperature via Henderson-Hasselbalch chemistry

TinyML on RP2040 (Embedded)

Fully on-device inference — no cloud, no internet required

N3IWF Local Edge Services

Central communication hub connecting Pico WH (MCU), RPi5 (MPU), and dashboard via local Wi-Fi — scalable for multiple devices without internet

Problem Statement

Existing intelligent monitoring methods fail to meet the simultaneous demands of adaptability, transparency, and embedded deployment in real-world aquaculture environments

01

No Adaptive + Transparent Embedded Monitoring

- Rule-based & Fuzzy Logic: transparent but static, cannot learn from data
- DQN: adaptive but opaque and too heavy for microcontrollers
- Hardware constraints of each device prevent a single device from handling all tasks

02

Existing RL Methods Require DO Sensors Despite pH-T Being Sufficient

- NH_3 toxicity, the true fish kill cause, is fully derivable from pH and temperature alone via Henderson-Hasselbalch
- Existing RL systems remain DO-centric despite DO being scientifically unnecessary for NH_3 estimation

03

No Embedded RL with Local Edge Monitoring

- All existing RL monitoring systems require cloud or server infrastructure
- No system uses a local edge hub enabling scalable multi-device deployment without internet

PROPOSED SOLUTION

Progressive Hybrid

Adaptive + Transparent + Embedded

pH + T Aware Reward

NH_3 toxicity derived directly from pH and temperature via Henderson-Hasselbalch, no DO sensor required

TinyML + N3IWF LES

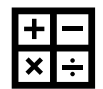
On-device inference + N3IWF as central communication hub scalable for multiple Pico WH devices



Research Objective



Develop a progressive hybrid FQL–DQN framework for aquaculture monitoring



Encode NH_3 toxicity via pH + temperature as the primary reward signal



Deploy the framework on TinyML embedded platform (RP2040)



Integrate N3IWF Local Edge Services as the central communication hub connecting Pico WH, RPi5, and dashboard enabling scalable multi-device deployment without internet



Literature Review

01 / SMART AQUACULTURE IoT

Flores-Iwasaki et al. (2025) — AgriEngineering, MDPI | Systematic Review & Bibliometric Analysis (2020–2024) | n = 217 papers

What They Did

Systematic review + bibliometric analysis of 217 IoT aquaculture papers (Scopus, 2020–2024). Mapped sensor usage, connectivity, and knowledge gaps across BFT, RAS, and aquaponic systems.

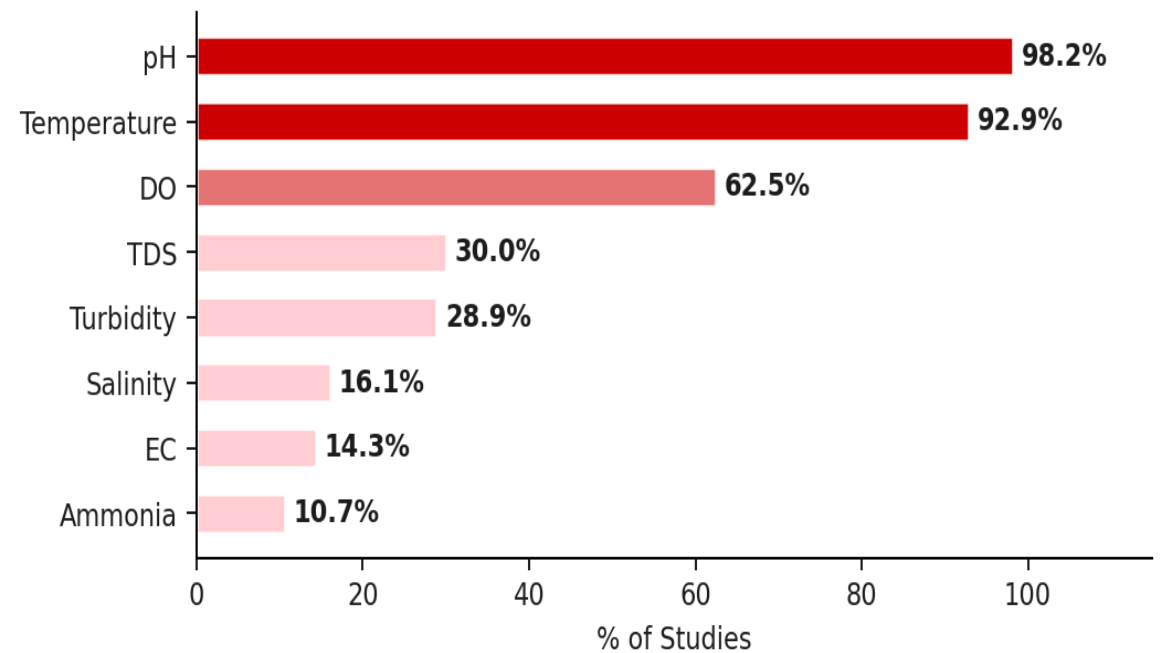
Key Output

pH monitored in 98.2% of studies, temperature in 92.9%. IoT monitoring improved growth rates, reduced mortality, and enabled rapid TAN detection. Wi-Fi and LoRaWAN are the dominant connectivity methods.

Gap Identified

No existing study uses pH + temperature as the sole input for adaptive embedded control. All reviewed works monitor only, none provide adaptive embedded monitoring on a microcontroller

IoT Sensor Usage in Aquaculture Studies (n=56)





Literature Review

02 / NH_3 TOXICITY IN AQUACULTURE

Lin et al. (2023) — *Water, MDPI*, vol. 15, no. 1 | doi: 10.3390/w15010027 | Review of acute NH_3 toxicity to aquatic decapod crustaceans

What They Did

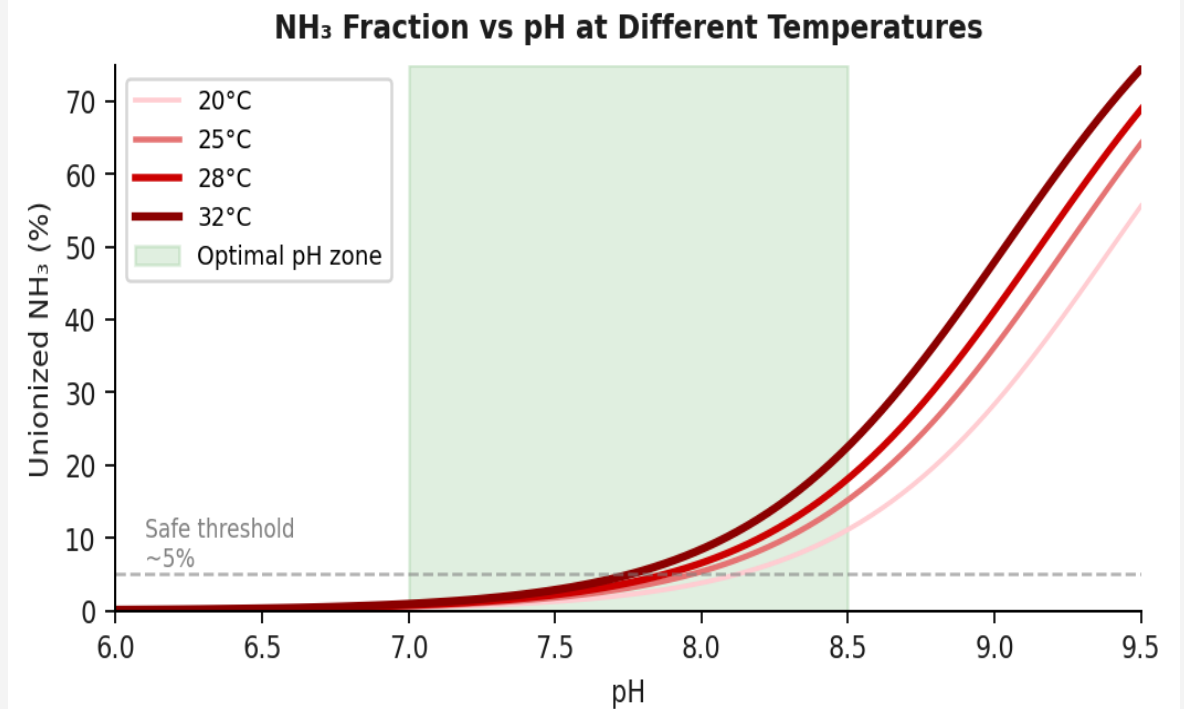
Comprehensive review of acute ammonia nitrogen ($\text{NH}_3\text{-N}$) toxicity in shrimp and crabs. Synthesized LC50 data across species, temperatures, pH levels, and salinity conditions.

Key Output

NH_3 toxicity is entirely governed by pH and temperature. Higher pH and temperature dramatically increase the toxic NH_3 fraction. At pH 8.5 & 30°C, NH_3 fraction rises $>5\times$ above safe threshold without any change in total ammonia concentration.

Gap Identified

no prior work encodes NH_3 toxicity risk directly as reward signal" → "no prior work encodes dual-sided pH biological risk — NH_3 toxicity via Henderson-Hasselbalch and acidosis via pH deviation — as a unified reward signal





Literature Review

03 / FUZZY Q-LEARNING FOR IoT EDGE SYSTEMS

Faraji-Mehmandar et al. (2023) — Journal of Ambient Intelligence and Humanized Computing, Springer | doi: 10.1007/s12652-023-04527-7

What They Did

Designed a Fuzzy Q-Learning (FQL) framework for autonomous resource provisioning of IoT applications in fog computing. Used MAPE-K control loop with FQL as the adaptive planning engine, evaluated on 3 workload scenarios.

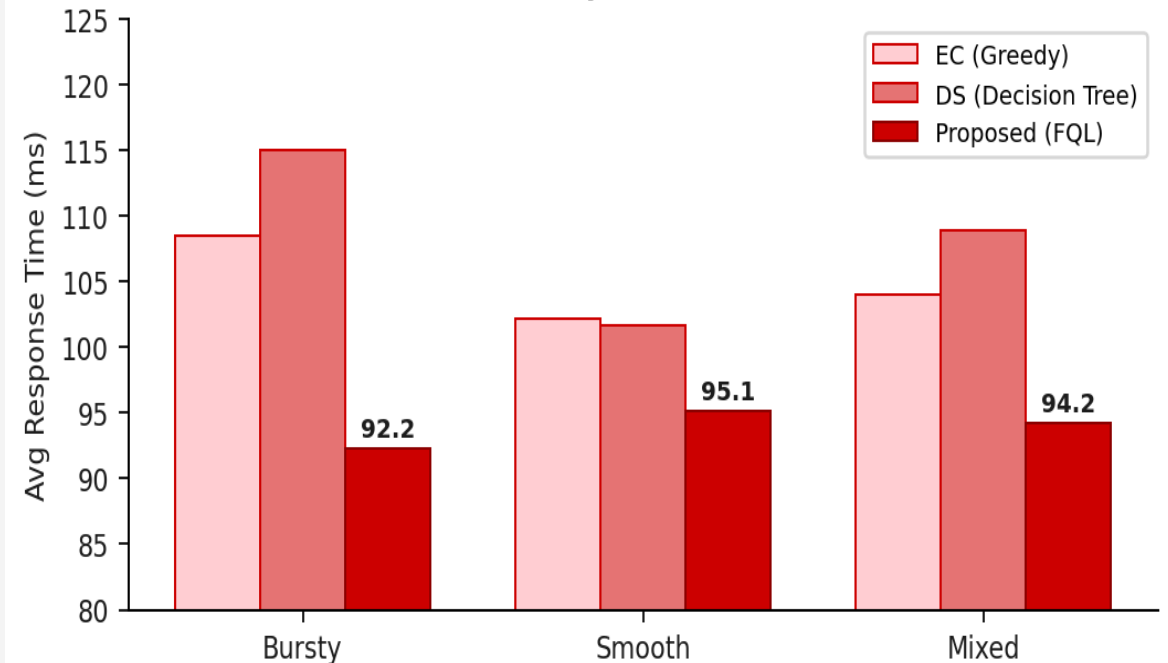
Key Output

FQL outperformed greedy and decision-tree baselines in all scenarios: response time reduced by ~15%, resource utilization improved to ~82%, scalability decisions more stable. FQL adapts without prior data rule initialization from domain knowledge is sufficient.

Gap Identified

FQL was applied to cloud/fog resource provisioning never to aquaculture monitoring. No work combines FQL with a biological reward signal (NH₃) or transitions from FQL to DQN progressively as

FQL vs Baselines: Response Time Across Scenarios





Literature Review

04 / DEEP Q-NETWORK (DQN) — FOUNDATIONAL RL METHOD

Mnih et al. (2015) — Nature, vol. 518, pp. 529–533 | doi: 10.1038/nature14236 | Google DeepMind

What They Did

Introduced DQN combining Q-learning with deep convolutional neural networks. Key innovations: experience replay (randomizes training samples) and a separate target network (stabilizes training). Tested on 49 Atari games from raw pixels.

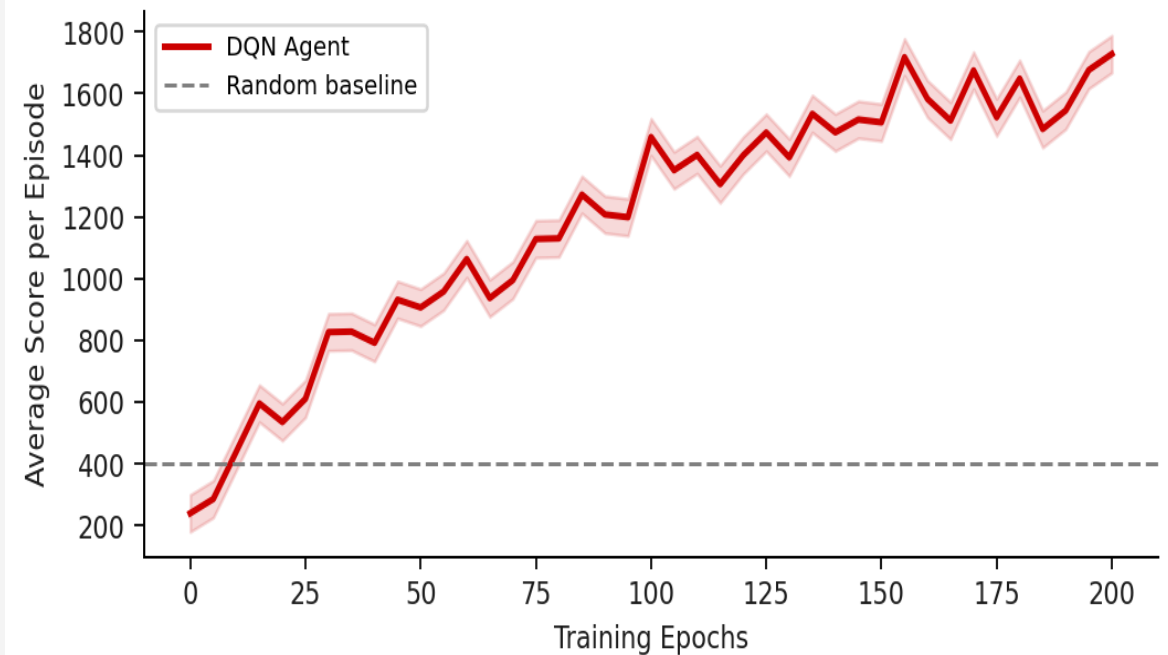
Key Output

DQN achieved human-level or above on 29/49 Atari games using the same architecture. Demonstrated end-to-end RL from high-dimensional inputs is feasible. Experience replay and target network are critical for stable convergence.

Gap Identified

DQN requires large experience data and high compute incompatible with TinyML (264 KB SRAM). No prior work deploys DQN on a microcontroller for environmental monitoring with pH+T-based NH_3 reward.

DQN Learning Curve — Avg Score vs Training Epochs





Literature Review

05 / LOCAL EDGE SERVICES via N3IWF IN 5G NETWORKS

Zampognaro et al. (2024) — IEEE MeditCom 2024 | doi: 10.1109/MeditCom61057.2024.10621141

What They Did

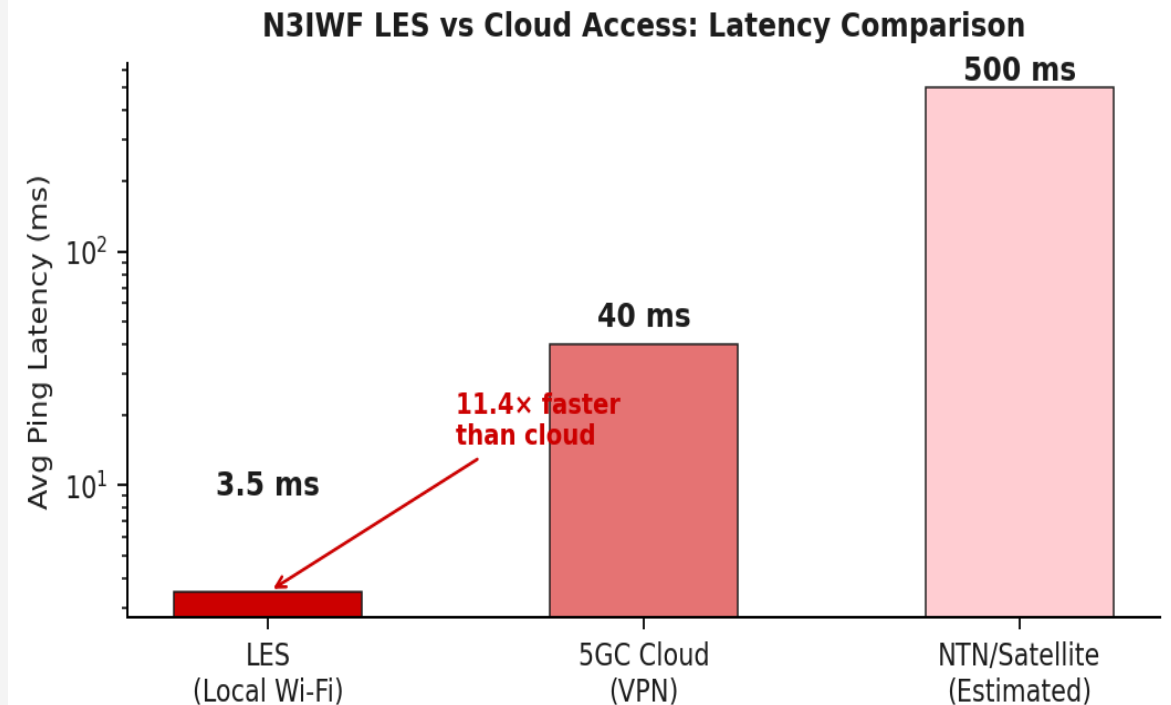
Proposed Local Edge Services (LES) architecture hosted within the N3IWF component of a 5G system. UEs connect via Wi-Fi and receive low-latency TCP/IP services at the edge without traversing the 5G Core. Validated with a real-time testbed.

Key Output

LES via N3IWF achieves 3.5 ms avg ping vs 40 ms via cloud 5GC (11.4× faster) and ~800 Mbit/s throughput vs 65.6 Mbit/s via cloud. QoS profiles from the 5G session are preserved for LES connections.

Gap Identified

LES was validated only for general IP services (CDN, DNS) never applied to IoT aquaculture monitoring. No prior work integrates N3IWF LES with an embedded TinyML device (RP2040 via Raspberry Pi) for remote farm dashboard.



Literature Comparison

Criteria	Mnih et al. (2015)	Faraji et al. (2023)	Zampognaro et al. (2024)	Flores-Iwasaki et al. (2025)	Lin et al. (2023)	Proposed System
Method	Deep Q-Network (DQN)	Fuzzy Q-Learning (FQL)	N3IWF Local Edge Services	IoT Sensor Monitoring Review	NH ₃ Toxicity Review	Progressive Hybrid FQL-DQN
Platform	GPU Cloud Server	Fog Computing Server	5G + Wi-Fi Testbed	Multi-platform IoT devices	Laboratory setup	RP2040 TinyML (264 KB SRAM)
Learning Type	Deep Reinforcement Learning	Fuzzy Reinforcement Learning	No learning	No learning	No learning	Progressive: Rule → FQL → DQN
Sensor Input	Game pixels (84×84)	Generic IoT workload	Generic network packets	pH, DO, T, EC, Turbidity, etc.	Lab chemical sensors	pH + Temperature only
Connectivity	Cloud dependent	Cloud/Fog dependent	5G Core + Wi-Fi	Cloud platforms (ThingSpeak, Blynk)	Not applicable	Wi-Fi Local (N3IWF)
Aquaculture Focus	No — Atari game domain	No — IoT resource management	No — network edge computing	Yes — water monitoring review	Yes — NH ₃ toxicity in fish	Yes — water quality monitoring
NH ₃ Awareness	Not applicable	Not applicable	Not applicable	Not considered	Theoretical only (review)	NH ₃ -aware reward via pH + T formula
Adaptive Monitoring	Game action control	Fog resource scaling	No control loop	Monitoring only — no actuator	No control	Aquaculture water quality control
Data Source	Atari simulator	iFogSim simulator	Real 5G testbed	217 papers (Scopus 2020–2024)	Literature review	Real field sensor data



Research Gap

Monitoring Without Adaptive Decision-Making

Source: Flores-Iwasaki et al. (2025)

98.2% of IoT aquaculture studies focus on sensing pH, DO, T. But automating adaptive responses remains a major challenge. No study implements an adaptive closed-loop monitoring system on embedded hardware.

Cloud-Dependent AI — Not Embeddable

Source: Mnih et al. (2015) · Faraji et al. (2023)

DQN requires GPU-scale compute; FQL runs on fog servers. No deep RL or fuzzy RL method has been deployed on a microcontroller (264 KB SRAM). Remote aquaculture sites cannot rely on cloud connectivity.

No Scalable Local Communication Layer

Source: Zampognaro et al. (2024)

Existing embedded systems use point-to-point USB connections limiting scalability to one device per deployment. No system uses a local edge hub (N3IWF) allowing multiple edge devices to coordinate without internet dependency.

No NH₃-Aware Reward Design

Source: Lin et al. (2023)

NH₃ toxicity is mathematically determined by pH and temperature (LC50: 1.0-3.6 mg/L). Yet no existing RL system uses this chemical relationship to define a biology-based reward function.

No Progressive Learning Strategy

Source: All reviewed studies

Existing systems start with full RL from day one risky without prior data. No study proposes a safe cold-start strategy that transitions Rule-Based to FQL to DQN progressively as real field data accumulates.



Theoretical Foundation — Fuzzy Q-Learning (FQL)

What is FQL?

Fuzzy Q-Learning combines Fuzzy Logic with Q-Learning (Reinforcement Learning). Instead of discrete states, FQL maps inputs to fuzzy membership functions, enabling gradient decision-making in noisy, continuous environments like aquaculture ponds.

Why FQL for Aquaculture?

- ✓ Zero cold-start data required, uses fuzzy domain rules
- ✓ Only 144 bytes, fits in 264 KB RP2040 SRAM
- ✓ Outperforms RL in stochastic IoT environments (Faraji 2023)
- ✓ Transparent: 25 IF-THEN fuzzy rules (pH x Temperature)

Update Rule (TD Learning):

$$Q(s,a) \leftarrow Q(s,a) + \alpha \cdot \mu_i \cdot [r + \gamma \cdot \max_{a'} Q(s') - Q(s,a)]$$

where μ_i = firing strength of active fuzzy rule i

FQL Rule Structure: 5 pH States × 5 Temperature States = 25 Rules

	Very Cold	Cold	Optimal	Hot	Very Hot
Very Acidic	HIGH	HIGH	HIGH	HIGH	HIGH
Acidic	MED	MED	MED	MED	MED
Normal	LOW	LOW	LOW	LOW	LOW
Alkaline	LOW	LOW	LOW	MED	MED
Very Alkaline	MED	MED	HIGH	HIGH	HIGH

LOW

<5% NH₃, pH safe

MED

5–15% NH₃, or pH 6.0–6.5

HIGH

>15% NH₃, or pH <6.0



Theoretical Foundation — Deep Q-Network (DQN)

What is DQN?

DQN (Mnih et al., 2015) uses a neural network to approximate Q-values for all actions simultaneously. Key innovations: Experience Replay (randomizes training samples) and Target Network (stabilizes convergence).

Why DQN — Stage 3?

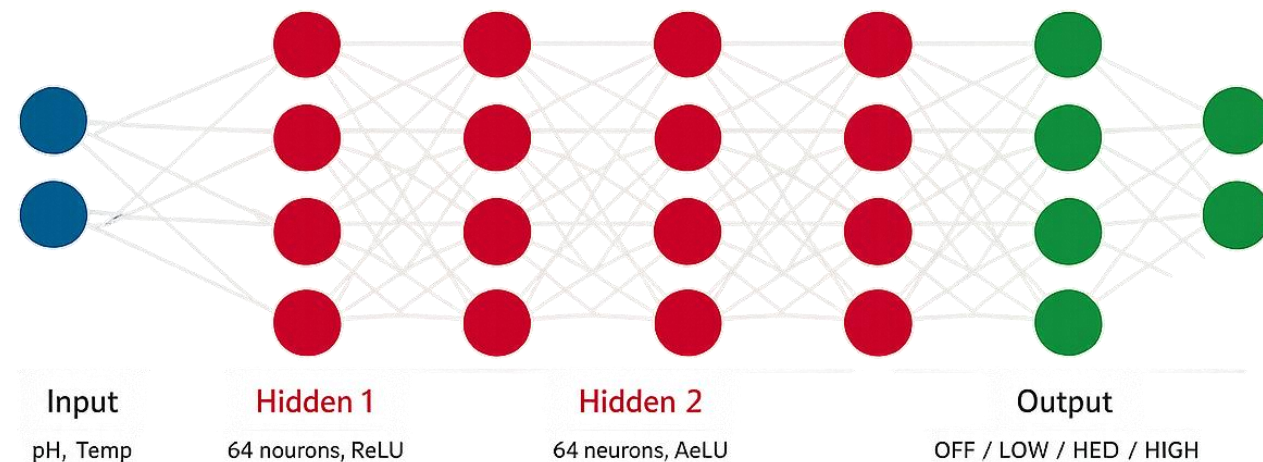
✓ Captures non-linear pH, Temperature, NH₃ relationships ✓ Trained on full replay buffer (real + virtual data) ✓ Policy distilled to Q-table (~5KB) for RP2040 inference ✓ Highest reward, outperforms Rule-Based and FQL

Bellman Update:

$$Q(s,a) \leftarrow r + \gamma \cdot \max_{a'} Q(s', a') \quad | \quad \text{Loss: } \text{MSE}[Q_{\text{target}} - Q_{\text{pred}}]$$

Architecture: Input(2) → Dense(64,ReLU) → Dense(64,ReLU) → Output(4)

DQN Architecture on Edge (Policy Distillation)



Policy Distillation for TinyML

DQN trains on RPi5 (PyTorch/NumPy). Evaluates Q-values on 2,500-state grid (50 pH × 50 Temp) → exported as lookup table (~5KB). RP2040 performs O(1) argmax, no matrix multiply needed.



Theoretical Foundation —N3IWF & Why This Research Needs It

What is N3IWF?

Non-3GPP Interworking Function (N3IWF) is a 5G Core component that allows non-radio devices (Wi-Fi, wired) to connect to the 5G network via IPsec IKEv2 tunnels — without requiring a 5G NR radio interface.

Ultra-Low Latency LES

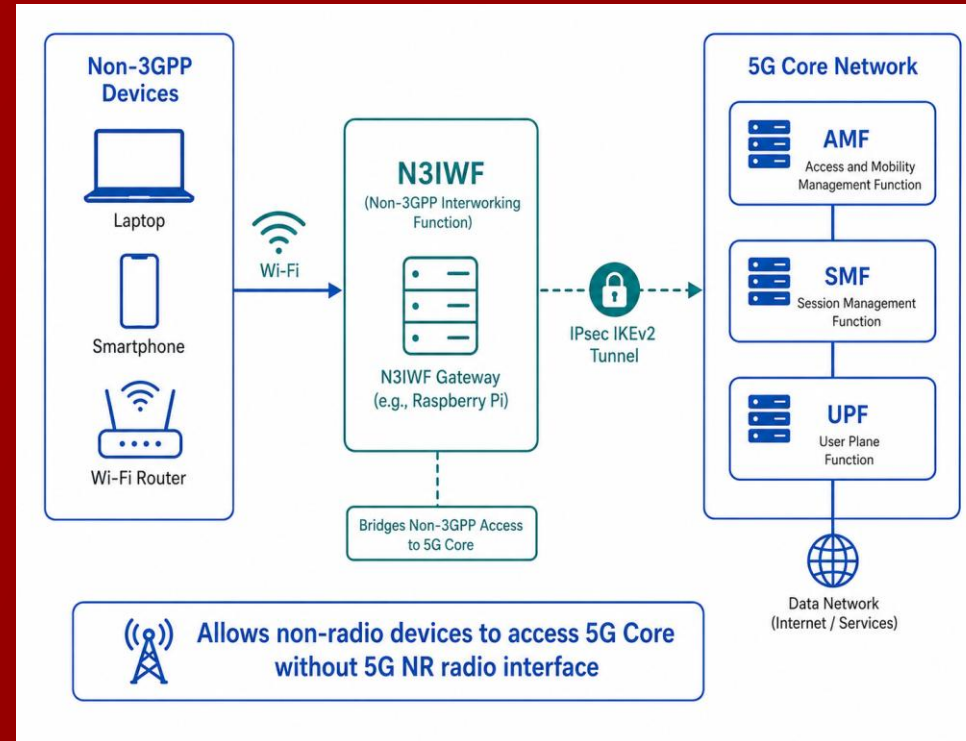
N3IWF hosts Local Edge Services (LES) directly at the gateway. Avg RTT: 3.5 ms vs 40 ms via cloud — 11.4× faster. Throughput: ~800 Mbit/s vs 65.6 Mbit/s via cloud 5GC. (Zampognaro et al., 2024 — IEEE MeditCom)

Why Essential for This Research

Aquaculture farms are in remote areas with unreliable internet. N3IWF enables: (1) Secure IPsec tunnel simulated on RPi5, (2) Direct TCP connection from Pico WH to RPi5, (3) Dashboard access without internet — all offline-capable.

Scalability Contribution

Unlike USB point-to-point connections, N3IWF LES acts as a central hub — multiple Pico WH devices can connect simultaneously via direct TCP connection.





Theoretical Foundation — NH_3 Toxicity & Reward Signal Design

Why pH + Temperature as the Primary Monitoring Signal?

NH_3 toxicity is the primary cause of invisible mass mortality in aquaculture and it is a direct mathematical function of pH and temperature alone (Henderson-Hasselbalch equation). No ammonia sensor needed. pH (98.2%) and temperature (92.9%) are already the most universally measured parameters in IoT aquaculture systems, making NH_3 inference practical and sensor-efficient.

NH_3 Dual-Sided pH Risk

- High pH + high temperature $\rightarrow$ high NH_3 fraction (Henderson-Hasselbalch)
- pH < 6.5 $\rightarrow$ direct acidosis risk independent of NH_3 fraction (EIFAC 1971)
- Both sides captured in unified r_{safety} reward component

Henderson-Hasselbalch Chemistry:

$$\text{NH}_3_{\text{fraction}} = 1 / (1 + 10^{(pK_a - \text{pH})})$$

$$pK_a(T) = 0.09018 + (2729.92 / (273.15 + T^\circ\text{C})) \quad | \quad \text{LC50: 1.0–3.6 mg/L (Lin et al. 2023)}$$

Reward Function: $r(t) = w_1 \cdot r_{\text{safety}} + w_2 \cdot r_{\text{stability}}$

r_{safety} [w = 0.8]

PRIMARY — Fish survival monitoring

$$r_{\text{safety}} = \alpha \times r_{\text{NH}_3} + \beta \times r_{\text{pH_dev}}$$

- r_{NH_3} : NH_3 fraction via Henderson-Hasselbalch
Safe <5%, Warning 5–15%, Danger >15% (Lin et al. 2023)
- $r_{\text{pH_dev}}$: piecewise acidosis penalty
pH < 6.0 $\rightarrow$ HIGH | pH 6.0–6.5 $\rightarrow$ MED | pH 6.5–8.5 $\rightarrow$ 0
(Boyd 1998, EIFAC 1971)

$r_{\text{stability}}$ [w = 0.2]

Smooth monitoring signal quality

$$r_{\text{stability}} = -\gamma \cdot |\Delta \text{pH}| - \delta \cdot |\Delta \text{Temp}|$$

Penalizes rapid pH and temperature fluctuations in the monitoring signal. Ensures reliable, stable readings for dashboard visualization and accurate NH_3 risk assessment.



Why Progressive + Hybrid?

Why this design is the only viable solution for embedded aquaculture monitoring

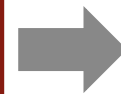
Stage 1
Rule-Based



Stage 2
FQL Learning



Stage 3
DQN Optimal



Proposed
System ✓

Why PROGRESSIVE?

Cold-Start Safety

DQN cannot train on live fish. Rule-Based ensures safe monitoring from Day 1

Data-Efficient Growth

FQL learns from real field data immediately. Q-values improve over time without large dataset.

Safe DQN Graduation

DQN activates only after $D \geq 10,000$ samples maximizing accuracy without early risk.

Why HYBRID?

DQN Alone Too Heavy

DQN needs GPU RAM. RP2040 has 264 KB SRAM. FQL = 144 bytes, the only RL that fits.

FQL Handles Uncertainty

Aquaculture is noisy & nonlinear. FQL outperforms pure RL in stochastic IoT environments.

Best of Both Combined

FQL = safe + interpretable early stage. DQN = long-term optimization. Together = all covered.

What We Get

Safety Guaranteed

Rule-Based hard constraints always active

Embedded Ready

DQN Q-Table: ~5KB (distilled policy) on 264KB SRAM

NH₃-Aware Monitoring

Reward based on pH + T → inferred NH₃ risk

Local Operation

Wi-Fi local via N3IWF → Simulated on RPi5



Methodology – System Architecture & Hardware

Hardware Components

RP2040 (TinyML Edge)

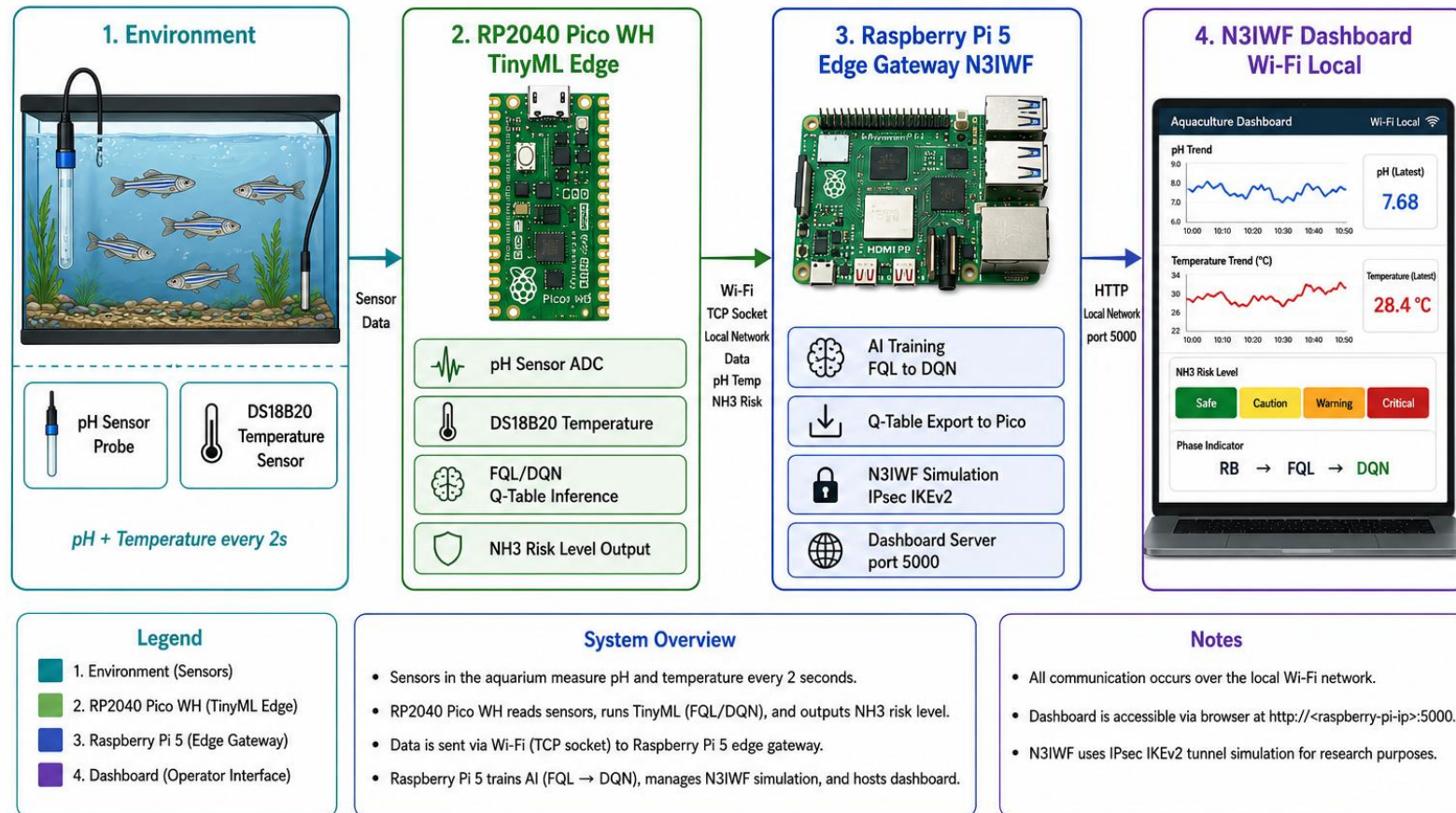
Dual-core Cortex-M0+, 264 KB SRAM
Runs FQL Q-table (25×4) or DQN Q-table policy
pH ADC read + DS18B20 1-Wire + Wi-Fi via N3IWF

Raspberry Pi 5 (Training Server)

Python + PyTorch training environment
FQL training → Q-table export via Wi-Fi N3IWF
DQN training → DQN Q-table export via Wi-Fi N3IWF

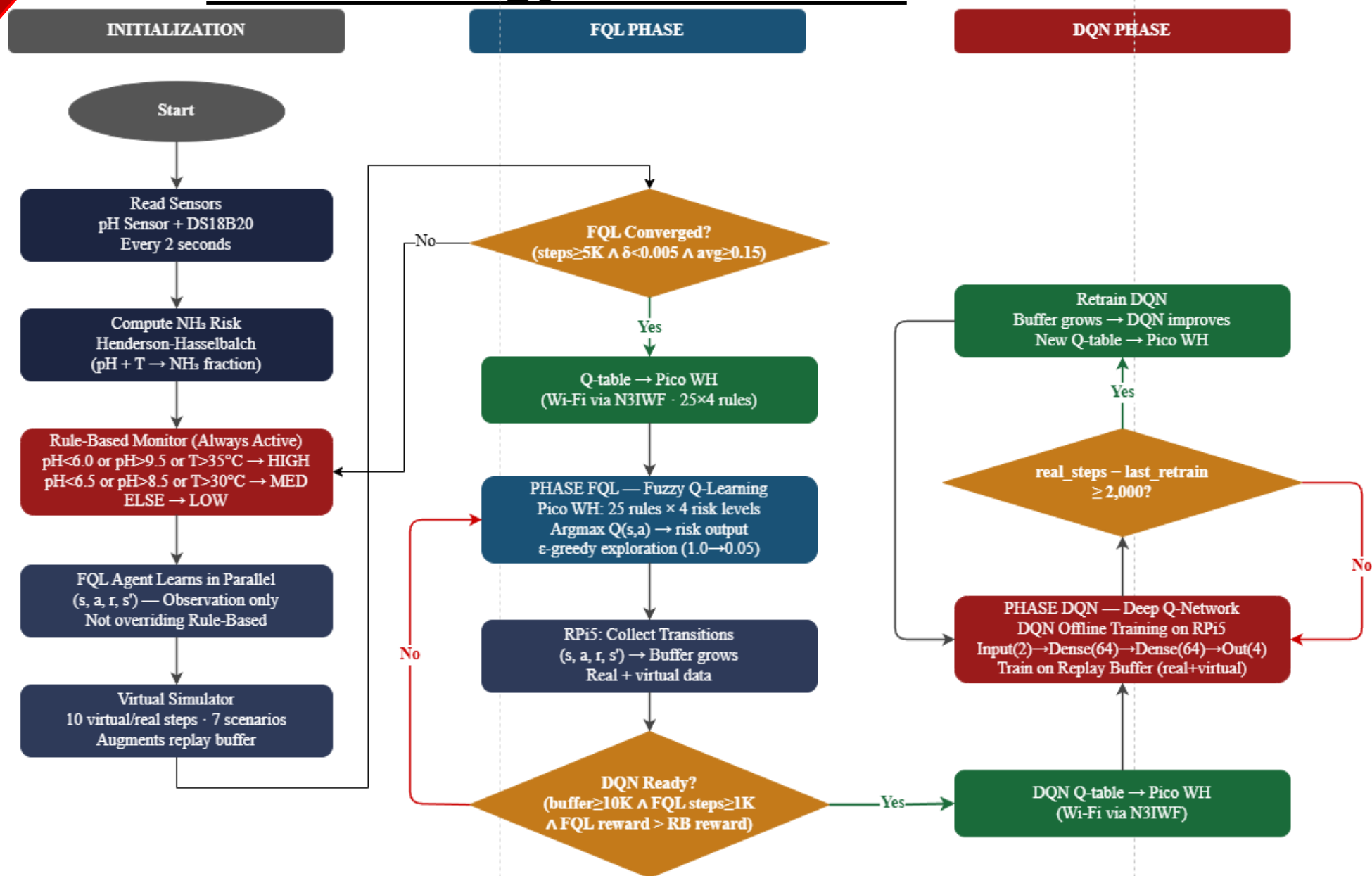
Sensors

pH Sensor (analog ADC) — range 0–14, ± 0.1 accuracy
DS18B20 Temperature Sensor — 1-Wire digital, $\pm 0.5^\circ$

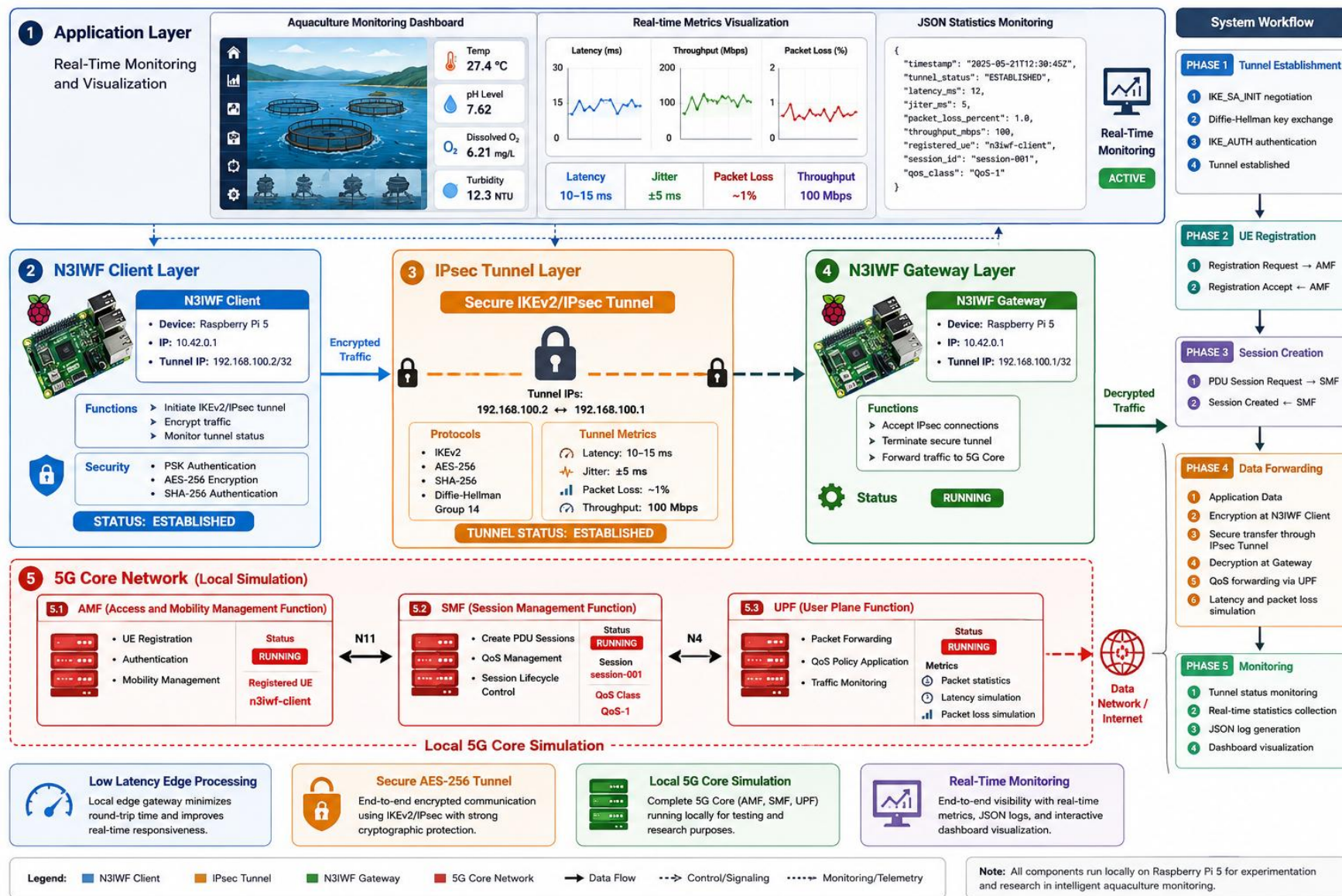




Methodology – Flowchart



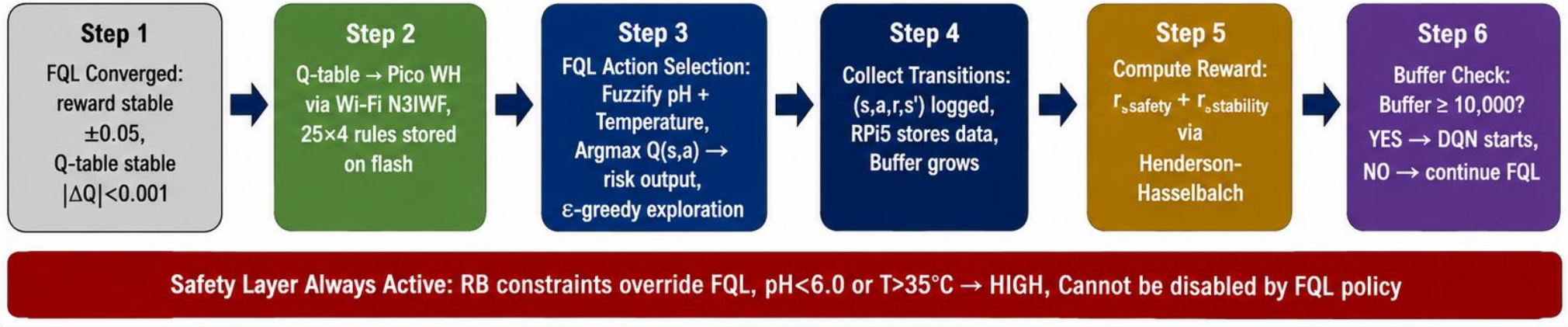
Methodology — N3IWF Local Edge Services



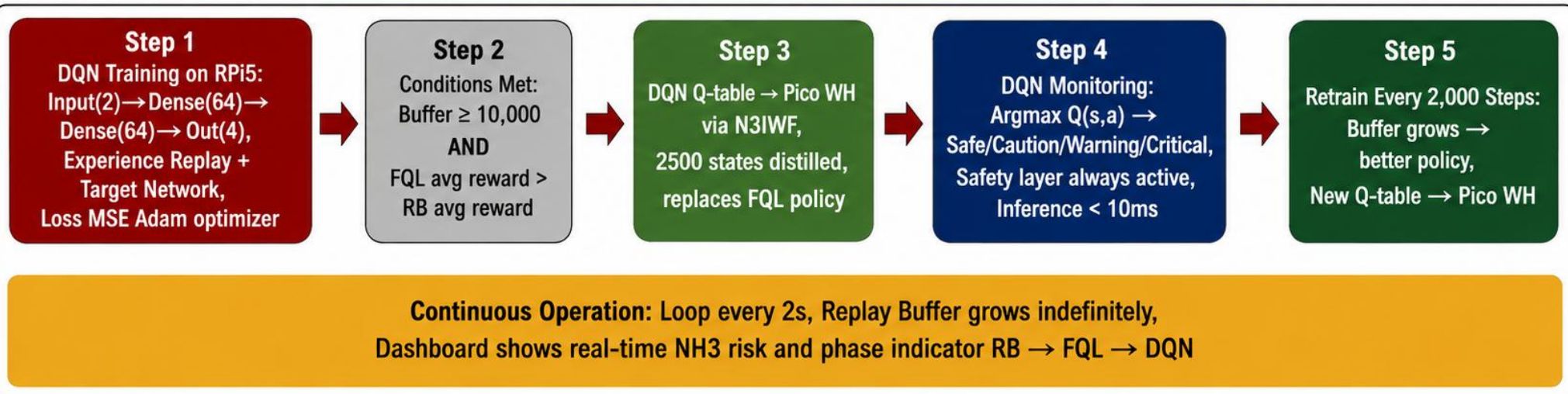


Methodology — Progressive Learning: FQL → DQN

PHASE FQL — Fuzzy Q-Learning



PHASE TRANSITION: Buffer $\geq 10,000$ AND FQL steps $\geq 1,000$ AND FQL reward $>$ RB reward





Expected Results

4 key contributions from this research

01 Progressive AI Convergence

- ▶ avg reward: DQN > FQL > Rule-Based (0.102 vs 0.102 vs 0.047)
- ▶ FQL converges via TD learning on real Pico 2W sensor data
- ▶ DQN retrained every 2,000 steps, policy improves continuously
- ▶ Visualized as reward convergence graph across all 3 phases

03 Sensor Validation vs IoT

- ▶ pH sensor (ADC linear calibration): error < 2% across 3 test conditions
- ▶ DS18B20 temperature (12-bit, $\pm 0.5^{\circ}\text{C}$): error < 1% vs reference
- ▶ Validates reward signal inputs, accurate pH+T = accurate NH_3 computation
- ▶ Inaccurate sensor $\rightarrow$ corrupt FQL/DQN training, validation is prerequisite

02 5G N3IWF Network Performance

- ▶ 100 ICMP ping packets: RTT, Jitter, PDR measured via IPsec IKEv2
- ▶ Low RTT $\rightarrow$ real-time sensor delivery, Low Jitter $\rightarrow$ stable Q-table sync
- ▶ $\text{PDR} \geq 98\%$ $\rightarrow$ reliable channel for continuous aquaculture monitoring
- ▶ Dashboard pings N3IWF every 3 seconds — latency shown live

04 Real-Time Monitoring Dashboard

- ▶ 6 auto-updating charts: pH, Temp, NH_3 risk level, inference time, N3IWF latency, convergence
- ▶ IPsec status, active phase (RB/FQL/DQN), ϵ value, replay buffer size, all live
- ▶ Reads from state.json — no port conflict with AI training process
- ▶ Proves all components (sensor $\rightarrow$ AI $\rightarrow$ 5G) integrated in one observable platform



THANK YOU

Questions & Discussion

*Adaptive Aquaculture Water Quality Monitoring via Zebrafish
Using Progressive Hybrid FQL-DQN with N3IWF on TinyML Embedded
Platform*